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CS 151 - NICE + NIST Practitioners

18 NOV 2025

Agenda

- Career Path
- Federal Work Force
- OPM Job Classification Handbook
- NIST SP 800-181r1 = NICE Framework
- NIST Internal Report = NIST IR 8355 - NICE Framework Competency Areas
- OPM Cybersecurity Codes Linked to the NICE Cybersecurity Workforce Framework
- Career Pathway = SSM 722
- Career Pathway = SCA 612
- Preparing a Job-Ready Cybersecurity Workforce = [Learning.CISA.gov](https://learning.cisa.gov)
- Bonus = The Big Dance = Job Offers
- Double Bonus = New Career Fields...Artificial Intelligence

Career Path

What is your plan? (*Keep it liquid so it can be modified as you move along*)

P = Purpose = what purpose does your plan serve?

L = Logic = Is your plan logical and realistic?

A = Action = What steps do you need to take in order to put your plan into action?

N = Numbers = Your plan should have an order by numbers, step 1, step 2, etc.
because plans have an order of operations.

- *Federal Work Force*
- The Federal Work Force lists all of their career roles and job opportunities on **USAJOBS.GOV**
- Create a profile, look for the **XX99 Career field opportunities** for entry level positions. These are roles where you can develop in the career field.
- Relocation will offer many more opportunities
- NOTE: the [CyberSeek](#) and the Cyber Heat Map for statistics and Job Opportunities

OPM Job Classification Handbook

- OPM Job Classification Handbook
- 2210 Career Field = 12 Career Fields “Currently” *See Next Three Slides*
- NOTE: the 2299 for “Student Trainee” entry level development opportunities as well as the Scholarship For Service (SFS) opportunities...perhaps your Master’s or Doctoral Degree could be paid for.

HANDBOOK OF OCCUPATIONAL GROUPS AND FAMILIES

TABLE OF CONTENTS

INTRODUCTION	3
GLOSSARY	3
GENERAL INFORMATION	5
CHANGES IN THIS EDITION OF THE HANDBOOK	7
PART I	12
OUTLINE OF POSITION CLASSIFICATION PLAN FOR WHITE COLLAR OCCUPATIONAL GROUPS AND SERIES	12
DEFINITIONS OF WHITE COLLAR OCCUPATIONAL GROUPS AND SERIES.....	18
0000 – MISCELLANEOUS OCCUPATIONS GROUP.....	18
0100 – SOCIAL SCIENCE, PSYCHOLOGY, AND WELFARE GROUP.....	25
0200 – HUMAN RESOURCES MANAGEMENT GROUP.....	32
0300 – GENERAL ADMINISTRATIVE, CLERICAL, AND OFFICE SERVICES GROUP.....	36
0400 – NATURAL RESOURCES MANAGEMENT AND BIOLOGICAL SCIENCES GROUP.....	43
0500 – ACCOUNTING AND BUDGET GROUP.....	50
0600 – MEDICAL, HOSPITAL, DENTAL, AND PUBLIC HEALTH GROUP.....	54
0700 – VETERINARY MEDICAL SCIENCE GROUP	66
0800 – ENGINEERING AND ARCHITECTURE GROUP.....	67
0900 – LEGAL AND KINDRED GROUP	75
1000 – INFORMATION AND ARTS GROUP.....	80
1100 – BUSINESS AND INDUSTRY GROUP.....	85
1200 – COPYRIGHT, PATENT, AND TRADEMARK GROUP.....	91
1300 – PHYSICAL SCIENCES GROUP	93
1400 – LIBRARY AND ARCHIVES GROUP	98
1500 – MATHEMATICAL SCIENCES GROUP.....	100
1600 – EQUIPMENT, FACILITIES, AND SERVICES GROUP.....	103
1700 – EDUCATION GROUP	105
1800 – INSPECTION, INVESTIGATION, ENFORCEMENT, AND COMPLIANCE GROUP.....	108
1900 – QUALITY ASSURANCE, INSPECTION, AND GRADING GROUP.....	115
2000 – SUPPLY GROUP	116
2100 – TRANSPORTATION GROUP.....	118
2200 – INFORMATION TECHNOLOGY GROUP.....	123
ALPHABETICAL LISTING OF WHITE COLLAR OCCUPATIONAL GROUPS AND SERIES	125
PART II.....	130
OUTLINE OF POSITION CLASSIFICATION PLAN FOR TRADE, CRAFT, OR LABOR JOB FAMILIES AND OCCUPATIONS.....	130
DEFINITIONS OF TRADE, CRAFT, OR LABOR FAMILIES AND OCCUPATIONS	
135 2500 – WIRE COMMUNICATIONS EQUIPMENT INSTALLATION AND MAINTENANCE FAMILY	135
2600 – ELECTRONIC EQUIPMENT INSTALLATION AND MAINTENANCE FAMILY	136
2800 – ELECTRICAL INSTALLATION AND MAINTENANCE FAMILY	138

Series in this group are: **2210 – [Information Technology Management Series](#)*****

This series covers two-grade interval administrative positions that manage, supervise, lead, administer, develop, deliver, and support information technology (IT) systems and services. This series covers only those positions for which the paramount requirement is knowledge of IT principles, concepts, and methods; e.g., data storage, software applications, networking.

Information technology refers to systems and services used in the automated acquisition, storage, manipulation, management, movement, control, display, switching, interchange, transmission, assurance, or reception of information. Information technology includes computers, network components, peripheral equipment, software, firmware, services, and related resources.

This series covers, but is not limited to, the following specialties:

Policy and Planning – Work that involves a wide range of IT management activities that typically extend and apply to an entire organization or major components of an organization. This includes strategic planning, capital planning and investment control, workforce planning, policy and standards development, resource management, knowledge management, architecture and infrastructure planning and management, auditing, and information security management.

Enterprise Architecture – Work that involves the analysis, planning, design, implementation, documentation, assessment, and management of the enterprise structural framework to align IT strategy, plans, and systems with the mission, goals, structure, and processes of the organization.

Security – Work that involves ensuring the confidentiality, integrity, and availability of systems, networks, and data through the planning, analysis, development, implementation, maintenance, and enhancement of information systems security programs, policies, procedures, and tools.

Systems Analysis – Work that involves applying analytical processes to the planning, design, and implementation of new and improved information systems to meet the business requirements of customer organizations.

Applications Software – Work that involves the design, documentation, development, modification, testing, installation, implementation, and support of new or existing applications software.

- Operating Systems** – Work that involves the planning, installation, configuration, testing, implementation, and management of the systems environment in support of the organization's IT architecture and business needs.
- Network Services** – Work that involves the planning, analysis, design, development, testing, quality assurance, configuration, installation, implementation, integration, maintenance, and/or management of networked systems used for the transmission of information in voice, data, and/or video formats.
- Data Management** – Work that involves the planning, development, implementation, and administration of systems for the acquisition, storage, and retrieval of data.
- Internet** – Work that involves the technical planning, design, development, testing, implementation, and management of Internet, intranet, and extranet activities, including systems/applications development and technical management of Web sites. This specialty only includes positions requiring the application of technical knowledge of Internet systems, services, and technologies.
- Systems Admin** = Work that involves planning and coordinating the installation, testing, operation, troubleshooting, and maintenance of hardware and software systems.
- Customer Support** – Work that involves the planning and delivery of customer support services, including installation, configuration, troubleshooting, customer assistance, and/or training, in response to customer requirements.

2299 – Information Technology Student Trainee Series

See the series definition for the General Student Trainee Series, 0099.

NIST Special Publication 800-181
Revision 1

Workforce Framework for Cybersecurity (NICE Framework)

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This publication is available free of charge from:
<https://doi.org/10.6028/NIST.SP.800-181r1>

NICE Framework

- National Initiative for Cybersecurity Education (NICE)
- Takes the current 12 specialties for OPM 2210 “Information Technology Management Series” and converts them into:
 - 7 Categories
 - 33 Specialty Areas
 - 52 Work Roles
- Work Roles focus on Tasks that are to be accomplished and establish proficiency in the duties to be performed. These work roles are used for Recruiting in Job Announcements (found on USAJOBS.GOV) and for Retention, when developing within the career field.
- The work roles will require the Knowledge, and Skills to perform the Tasks required for each position duty.
 - Knowledge gives and understanding of what the role is and WHAT must be known
 - Skills create the practical application of HOW to perform the duties.

NIST Internal Report = NIST IR 8355 - NICE Framework Competency Areas

**NIST Internal Report
NIST IR 8355**

NICE Framework Competency Areas

Preparing a Job-Ready Cybersecurity Workforce

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This publication is available free of charge from:
<https://doi.org/10.6028/NIST.IR.8355>

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U.S. Department of Commerce
Gina M. Raimondo, Secretary

National Institute of Standards and Technology
Laurie E. Locascio, NIST Director and Under Secretary of Commerce for Standards and Technology

NIST Internal Report = NIST IR 8355 - NICE Framework Competency Areas

Abstract

This publication from NICE describes Competency Areas as included in the *Workforce Framework for Cybersecurity (NICE Framework)*, NIST Special Publication 800-181, Revision 1. The NICE Framework is a fundamental reference for describing and sharing information about cybersecurity work.

At its core are Task, Knowledge, and Skill (TKS) statements that provide a foundation for learners, including students, job seekers, and employees.

These statements are then used to define Competency Areas—clusters of related Knowledge and Skill statements that correlates with one’s capability to perform Tasks in a particular domain—and Work Roles, which are composed of Task statements that constitute work for which someone is responsible or accountable.

This document shares more detail about what NICE Framework Competency Areas are, including their evolution and development, clarifies the differences between Competency Areas and Work Roles, and provides example uses for Competency Areas from various stakeholder perspectives, and. Finally, the publication identifies where the NICE Framework list of Competency Areas is separately published and maintained distinct from this publication so it can be updated more frequently as a flexible and contemporary reference resource.

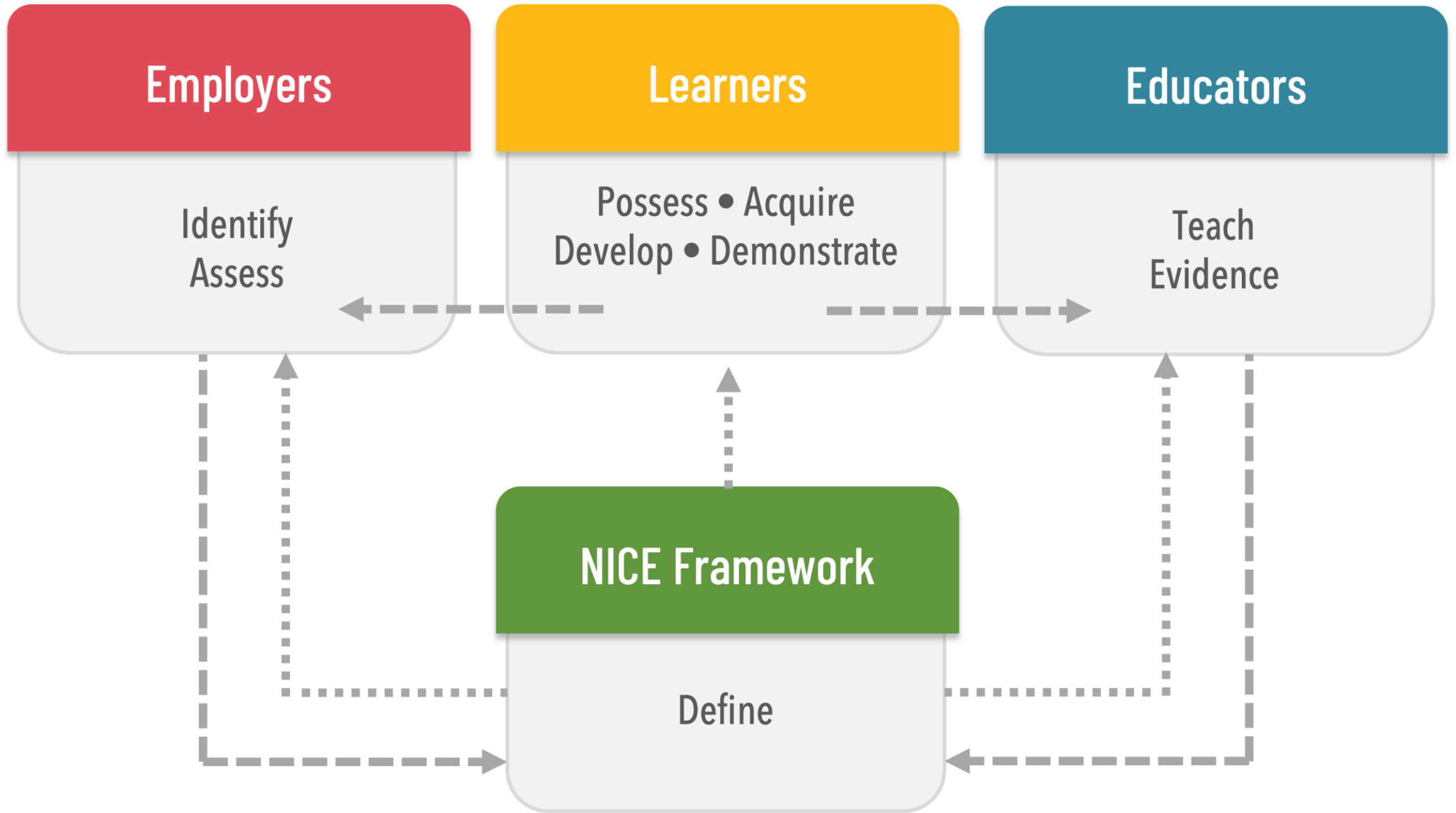
Audience

The NICE Framework serves as a bridge between employers and education, training, and certification providers as well as a tool to help learners determine needs and demonstrate capabilities.

A standardized approach to Competency Areas provides accessible information about what individuals need to know and be able to do, enables the development of more effective learning, and establishes regular processes to consistently describe and validate a learner's capabilities.

Therefore, the following are both stakeholders and the audience for this work: employers; workforce development and human resources professionals; education, training, and certification providers; and learners.

NIST Internal Report = NIST IR 8355 - NICE Framework Competency Areas



NIST Internal Report = NIST IR 8355 - NICE Framework Competency Areas

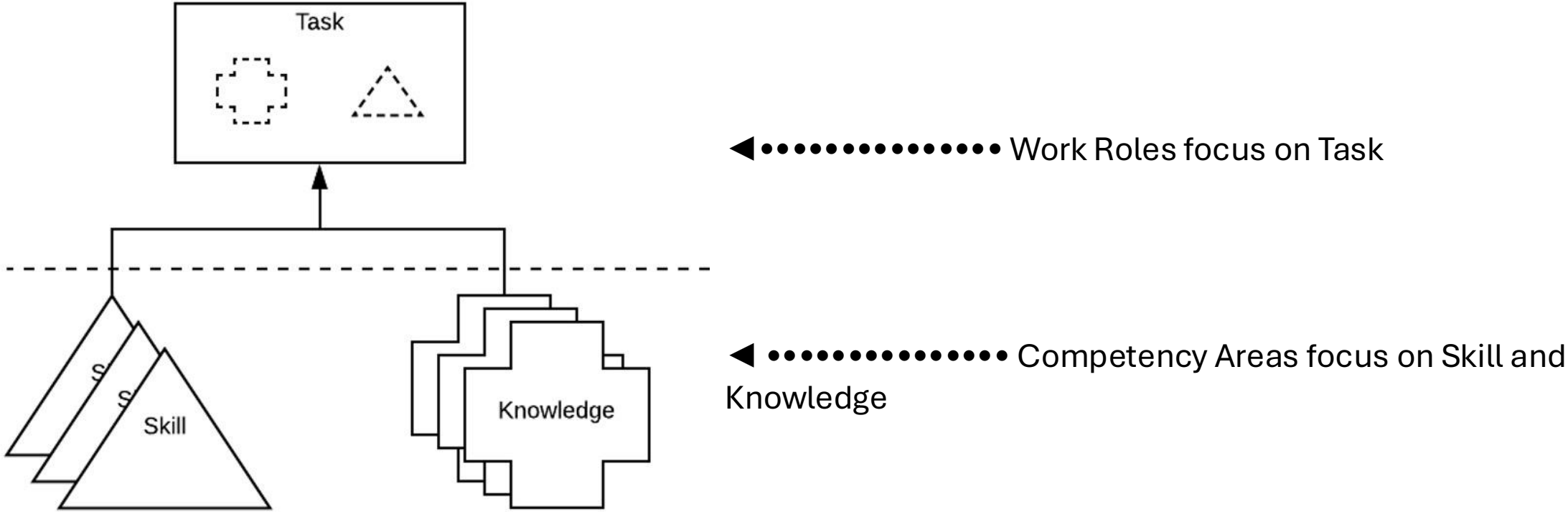


Table 1. NICE Framework Competency Areas vs. Work Roles

Competency Areas	Work Roles
Learner focused	Work focused
Help address employer needs	Help define positions and responsibilities
Assessment is typically based on a Competency Area as a whole	Assessment typically occurs at the Task level

Competency Areas help address employers’ capability needs more broadly, while Work Roles are used when defining specific positions and responsibilities. Finally, assessment is typically based on the Competency Area as a whole, whereas assessment for Work Roles typically occurs at the Task level.

OPM Cybersecurity Codes Linked to the NICE Cybersecurity Workforce Framework

NOTE the OPM Code for the work role code

OPM Cybersecurity Codes Linked to the NICE Cybersecurity Workforce Framework

Table 1: Work Role Descriptions and New Cybersecurity Codes

Category	Specialty Area	Work Role	OPM Code	Work Role Description
Securely Provision	Risk Management	Authorizing Official/Designating Representative	611	Senior official or executive with the authority to formally assume responsibility for operating an information system at an acceptable level of risk to organizational operations (including mission, functions, image, or reputation), organizational assets, individuals, other organizations, and the Nation (CNSSI 4009).
		Security Control Assessor	612	Conducts independent comprehensive assessments of the management, operational, and technical security controls and control enhancements employed within or inherited by an information technology (IT) system to determine the overall effectiveness of the controls (as defined in NIST SP 800-37).
	Software Development	Software Developer	621	Develops, creates, maintains, and writes/codes new (or modifies existing) computer applications, software, or specialized utility programs.
		Secure Software Assessor	622	Analyzes the security of new or existing computer applications, software, or specialized utility programs and provides actionable results.
	Systems Architecture	Enterprise Architect	651	Develops and maintains business, systems, and information processes to support enterprise mission needs; develops information technology (IT) rules and requirements that describe baseline and target architectures.
		Security Architect	652	Designs enterprise and systems security throughout the development life cycle; translates technology and environmental conditions (e.g., law and regulation) into security designs and processes.

Category	Specialty Area	Work Role	OPM Code	Work Role Description
	Technology R&D	Research & Development Specialist	661	Conducts software and systems engineering and software systems research in order to develop new capabilities, ensuring cybersecurity is fully integrated. Conducts comprehensive technology research to evaluate potential vulnerabilities in cyberspace systems.
	Systems Requirements Planning	Systems Requirements Planner	641	Consults with customers to evaluate functional requirements and translate functional requirements into technical solutions.
	Test and Evaluation	System Testing and Evaluation Specialist	671	Plans, prepares, and executes tests of systems to evaluate results against specifications and requirements as well as analyze/report test results.
	Systems Development	Information Systems Security Developer	631	Designs, develops, tests, and evaluates information system security throughout the systems development life cycle.
		Systems Developer	632	Designs, develops, tests, and evaluates information systems throughout the systems development life cycle.
Operate and Maintain	Data Administration	Database Administrator	421	Administers databases and/or data management systems that allow for the storage, query, and utilization of data.
		Data Analyst	422	Examines data from multiple disparate sources with the goal of providing new insight. Designs and implements custom algorithms, flow processes, and layouts for complex, enterprise-scale data sets used for modeling, data mining, and research purposes.
	Knowledge Management	Knowledge Manager	431	Responsible for the management and administration of processes and tools that enable the organization to identify, document, and access intellectual capital and information content.
	Customer Service and Technical Support	Technical Support Specialist	411	Provides technical support to customers who need assistance
				utilizing client-level hardware and software in accordance with established or approved organizational process components. (i.e., Master Incident Management Plan, when applicable).

Category	Specialty Area	Work Role	OPM Code	Work Role Description
	Network Services	Network Operations Specialist	441	Plans, implements, and operates network services/systems, to include hardware and virtual environments.
	Systems Administration	System Administrator	451	Installs, configures, troubleshoots, and maintains hardware and software, and administers system accounts.
	Systems Analysis	Systems Security Analyst	461	Responsible for the analysis and development of the integration, testing, operations, and maintenance of systems security.
Oversee and Govern	Legal Advice and Advocacy	Cyber Legal Advisor	731	Provides legal advice and recommendations on relevant topics related to cyber law.
		Privacy Compliance Manager	732	Develops and oversees privacy compliance program and privacy program staff, supporting privacy compliance needs of privacy and security executives and their teams.
	Training, Education, and Awareness	Cyber Instructional Curriculum Developer	711	Develops, plans, coordinates, and evaluates cyber training/education courses, methods, and techniques based on instructional needs.
		Cyber Instructor	712	Develops and conducts training or education of personnel within cyber domain.
	Cybersecurity Management	Information Systems Security Manager	722	Responsible for the cybersecurity of a program, organization, system, or enclave.
		COMSEC Manager	723	Manages the Communications Security (COMSEC) resources of an organization (CNSSI 4009).
	Strategic Planning and Policy	Cyber Workforce Developer and Manager	751	Develops cyberspace workforce plans, strategies and guidance to support cyberspace workforce manpower, personnel, training and education requirements and to address changes to cyberspace policy, doctrine, materiel, force structure, and education and training requirements.
		Cyber Policy and Strategy Planner	752	Develops cyberspace plans, strategy and policy to support and align with organizational cyberspace missions and initiatives.

Category	Specialty Area	Work Role	OPM Code	Work Role Description
	Executive Cyber Leadership	Executive Cyber Leadership	901	Executes decision-making authorities and establishes vision and direction for an organization's cyber and cyber-related resources and/or operations.
	Acquisition and Program/Project Management	Program Manager	801	Leads, coordinates, communicates, integrates and is accountable for the overall success of the program, ensuring alignment with critical agency priorities.
		IT Project Manager	802	Directly manages information technology projects to provide a unique service or product.
		Product Support Manager	803	Manages the package of support functions required to field and maintain the readiness and operational capability of systems and components.
		IT Investment/Portfolio Manager	804	Manages a portfolio of IT capabilities that align with the overall needs of mission and business enterprise priorities.
		IT Program Auditor	805	Conducts evaluations of an IT program or its individual components, to determine compliance with published standards.
Protect and Defend	Cyber Defense Analysis	Cyber Defense Analyst	511	Uses data collected from a variety of cyber defense tools (e.g., IDS alerts, firewalls, network traffic logs) to analyze events that occur within their environments for the purposes of mitigating threats.
	Cyber Defense Infrastructure Support	Cyber Defense Infrastructure Support Specialist	521	Tests, implements, deploys, maintains, and administers the infrastructure hardware and software.
	Incident Response	Cyber Defense Incident Responder	531	Investigates, analyzes, and responds to cyber incidents within the network environment or enclave.
	Vulnerability Assessment and Management	Vulnerability Assessment Analyst	541	Performs assessments of systems and networks within the network environment or enclave and identifies where those systems/networks deviate from acceptable configurations, enclave policy, or local policy. Measures effectiveness of defense-in-depth architecture against known vulnerabilities.

Category	Specialty Area	Work Role	OPM Code	Work Role Description
Analyze	Threat Analysis	Warning Analyst	141	Develops unique cyber indicators to maintain constant awareness of the status of the highly dynamic operating environment. Collects, processes, analyzes, and disseminates cyber warning assessments.
	Exploitation Analysis	Exploitation Analyst	121	Collaborates to identify access and collection gaps that can be satisfied through cyber collection and/or preparation activities. Leverages all authorized resources and analytic techniques to penetrate targeted networks.
	All-Source Analysis	All-Source Analyst	111	Analyzes data/information from one or multiple sources to conduct preparation of the environment, respond to requests for information, and submit intelligence collection and production requirements in support of planning and operations.
		Mission Assessment Specialist	112	Develops assessment plans and measures of performance/effectiveness. Conducts strategic and operational effectiveness assessments as required for cyber events. Determines whether systems performed as expected and provides input to the determination of operational effectiveness.
	Targets	Target Developer	131	Performs target system analysis, builds and/or maintains electronic target folders to include inputs from environment preparation, and/or internal or external intelligence sources. Coordinates with partner target activities and intelligence organizations, and presents candidate targets for vetting and validation.
		Target Network Analyst	132	Conducts advanced analysis of collection and open-source data to ensure target continuity; to profile targets and their activities; and develop techniques to gain more target information. Determines how targets communicate, move, operate and live based on knowledge of target technologies, digital networks and the applications on them.

Category	Specialty Area	Work Role	OPM Code	Work Role Description
	Language Analysis	Multi-Disciplined Language Analyst	151	Applies language and culture expertise with target/threat and technical knowledge to process, analyze, and/or disseminate intelligence information derived from language, voice and/or graphic material. Creates, and maintains language specific databases and working aids to support cyber action execution and ensure critical knowledge sharing. Provides subject matter expertise in foreign language-intensive or interdisciplinary projects.
Collect and Operate	Collection Operations	All Source-Collection Manager	311	Identifies collection authorities and environment; incorporates priority information requirements into collection management; develops concepts to meet leadership's intent. Determines capabilities of available collection assets, identifies new collection capabilities; and constructs and disseminates collection plans. Monitors execution of tasked collection to ensure effective execution of the collection plan.
		All Source-Collection Requirements Manager	312	Evaluates collection operations and develops effects-based collection requirements strategies using available sources and methods to improve collection. Develops, processes, validates, and coordinates submission of collection requirements. Evaluates performance of collection assets and collection operations.
	Cyber Operational Planning	Cyber Intel Planner	331	Develops detailed intelligence plans to satisfy cyber operations requirements. Collaborates with cyber operations planners to identify, validate, and levy requirements for collection and analysis. Participates in targeting selection, validation, synchronization, and execution of cyber actions. Synchronizes intelligence activities to support organization objectives in cyberspace.

Category	Specialty Area	Work Role	OPM Code	Work Role Description
		Cyber Ops Planner	332	Develops detailed plans for the conduct or support of the applicable range of cyber operations through collaboration with other planners, operators and/or analysts. Participates in targeting selection, validation, synchronization, and enables integration during the execution of cyber actions.
		Partner Integration Planner	333	Works to advance cooperation across organizational or national borders between cyber operations partners. Aids the integration of partner cyber teams by providing guidance, resources, and collaboration to develop best practices and facilitate organizational support for achieving objectives in integrated cyber actions.
	Cyber Operations	Cyber Operator	321	Conducts collection, processing, and/or geolocation of systems in order to exploit, locate, and/or track targets of interest. Performs network navigation, tactical forensic analysis, and, when directed, executing on-net operations.
Investigate	Cyber Investigation	Cyber Crime Investigator	221	Identifies, collects, examines, and preserves evidence using controlled and documented analytical and investigative techniques.
	Digital Forensics	Forensics Analyst	211	Conducts deep-dive investigations on computer-based crimes establishing documentary or physical evidence, to include digital media and logs associated with cyber intrusion incidents.
		Cyber Defense Forensics Analyst	212	Analyzes digital evidence and investigates computer security incidents to derive useful information in support of system/network vulnerability mitigation.
Not Applicable	Not Applicable	Not Applicable	000	Does NOT involve work functions in information technology (IT), cybersecurity, or cyber-related areas.

Preparing a Job-Ready Cybersecurity Workforce

[NICE Releases NICE Framework Components v2.0.0 | NIST](#)

[NICE Framework Components v2.0.0.xlsx](#)

Preparing a Job-Ready Cybersecurity Workforce

Work Role	Work Role Description	Work Role ID	OPM Code (Fed Use)	Link to TKS List
OVERSIGHT and GOVERNANCE (OG) – Provides leadership, management, direction, and advocacy so the organization may effectively manage cybersecurity-related risks to the enterprise and conduct cybersecurity work.				
Systems Security Management	Responsible for managing the cybersecurity of a program, organization, system, or enclave.	OG-WRL-014	722	Click to view OG-WRL-014 TKS List

Preparing a Job-Ready Cybersecurity Workforce

Work Role	Work Role Description	Work Role ID	OPM Code (Fed Use)	Link to TKS List
OVERSIGHT and GOVERNANCE (OG) – Provides leadership, management, direction, and advocacy so the organization may effectively manage cybersecurity-related risks to the enterprise and conduct cybersecurity work.				
Security Control Assessment	Responsible for conducting independent comprehensive assessments of management, operational, and technical security controls and control enhancements employed within or inherited by a system to determine their overall effectiveness.	OG-WRL-012	612	Click to view OG-WRL-012 TKS List

Preparing a Job-Ready Cybersecurity Workforce = Learning.CISA.gov

- <https://learning.cisa.gov/>
- You will have to set up an account on the CISA site,
but will also have to create an account on the **CISA OKTA Partner platform**

Bonus...*The Big Dance* = Job Offers

It's not just about the pay, think steps for experience, always ask for more, you can go down, but you can't go up on the requested pay amount. You may not get it, but ASK!

Benefits = Leave, both annual and medical, Flex schedule, etc.

What About Other Career Fields in Cybersecurity Not Yet Identified such as Artificial Intelligence?

Security for AI Fundamentals